



TRIOS CYBER

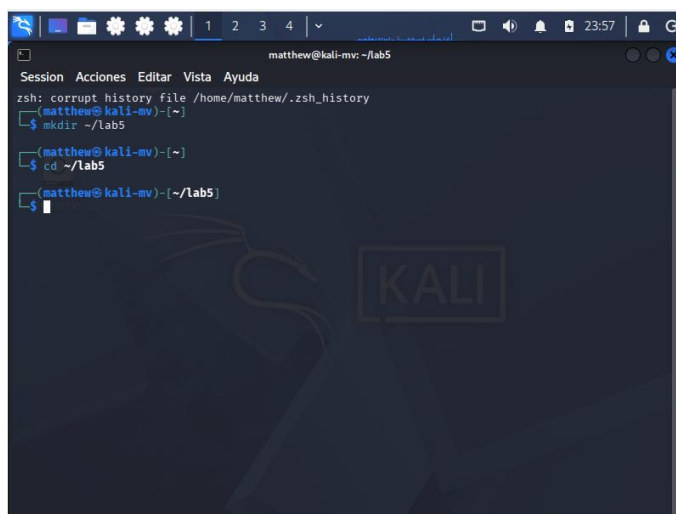
LABORATORY PRACTICE 5

Endpoint Baseline Collection and System State Documentation

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1. Baseline Workspace Preparation

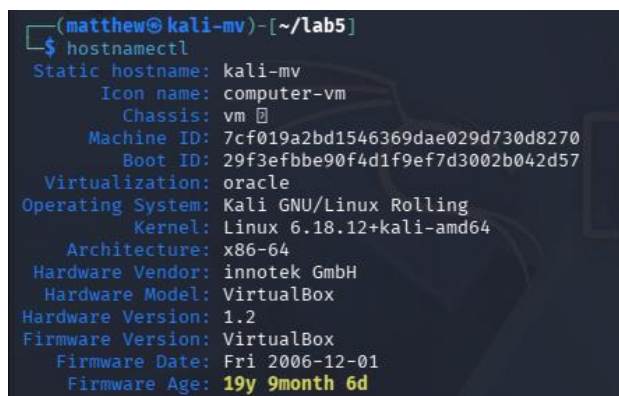
A dedicated lab5 directory was created on the Kali Linux endpoint to store the system baseline evidence and command outputs. The directory provided a single location for the files that would later be used during comparison and alert investigation.

A terminal window titled 'matthew@kali-mv: ~/lab5' showing the execution of commands to create and navigate into a directory named 'lab5'. The background features a Kali Linux logo.

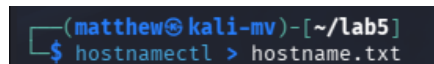
```
matthew@kali-mv: ~/lab5
Session Acciones Editar Vista Ayuda
zsh: corrupt history file /home/matthew/.zsh_history
(matthew@kali-mv)-[~]
$ mkdir ~/lab5
(matthew@kali-mv)-[~]
$ cd ~/lab5
(matthew@kali-mv)-[~/lab5]
$
```

2. Hostname Baseline Collection

The endpoint identity was collected with `hostnamectl`. The system reported the static hostname `kali-mv` and Kali GNU/Linux Rolling running as a VirtualBox virtual machine. The hostname information was then saved to `hostname.txt` for later comparison.

A terminal window showing the output of the 'hostnamectl' command. The output lists system details such as static hostname, icon name, chassis, machine ID, boot ID, virtualization type, operating system, kernel, architecture, hardware vendor, hardware model, hardware version, firmware version, firmware date, and firmware age.

```
(matthew@kali-mv)-[~/lab5]
$ hostnamectl
Static hostname: kali-mv
Icon name: computer-vm
Chassis: vm
Machine ID: 7cf019a2bd1546369dae029d730d8270
Boot ID: 29f3efbbe90f4d1f9ef7d3002b042d57
Virtualization: oracle
Operating System: Kali GNU/Linux Rolling
Kernel: Linux 6.18.12+kali-amd64
Architecture: x86_64
Hardware Vendor: innotek GmbH
Hardware Model: VirtualBox
Hardware Version: 1.2
Firmware Version: VirtualBox
Firmware Date: Fri 2006-12-01
Firmware Age: 19y 9month 6d
```

A terminal snippet showing the command to save the output of 'hostnamectl' to a file named 'hostname.txt'.

```
(matthew@kali-mv)-[~/lab5]
$ hostnamectl > hostname.txt
```

3. Current User Baseline Collection

The current user and account identity were recorded using whoami and id. The active laboratory user was matthew, and the account and group information were captured as part of the endpoint baseline. The results were saved to current_user.txt.

```
(matthew@kali-mv)-[~/lab5]
$ whoami
matthew

(matthew@kali-mv)-[~/lab5]
$ id
uid=1000(matthew) gid=1000(matthew) grupos=1000(matthew),4(adm),20(dialout),24(cdrom),25(floppy),27(sudo),29(audio),30(dip),44(video),46(plugdev),100(users),101(netdev),103(scanner),116(blue tooth),121(lpadmin),124(wireshark),130(vboxsf),131(kaboxer)
```

```
(matthew@kali-mv)-[~/lab5]
$ whoami > current_user.txt

(matthew@kali-mv)-[~/lab5]
$ id >> current_user.txt
```

4. Running Process Baseline

The running process state was collected with ps aux. The full process list was reviewed, while the first twenty entries were also displayed as a concise snapshot. The complete output was saved to running_processes.txt so that future investigations can identify new, missing, or unusual processes.

```
(matthew@kali-mv)-[~/lab5]
$ ps aux | head -20
USER          PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root             1  0.0  0.3 24628 15528 ?        Ss   sep06 0:01 /sbin/init splash
root             2  0.0  0.0      0     0 ?        S    sep06 0:00 [kthreadd]
root             3  0.0  0.0      0     0 ?        S    sep06 0:00 [pool_workqueue_release]
root             4  0.0  0.0      0     0 ?        I<   sep06 0:00 [kworker/R-rcu_gp]
root            10  0.0  0.0      0     0 ?        I<   sep06 0:00 [kworker/R-sync_wq]
root            11  0.0  0.0      0     0 ?        I<   sep06 0:00 [kworker/R-kvfree_rcu_reclai
0]
root             7  0.0  0.0      0     0 ?        I<   sep06 0:00 [kworker/R-slub_flushwq]
root             8  0.0  0.0      0     0 ?        I<   sep06 0:00 [kworker/R-netns]
root            10  0.0  0.0      0     0 ?        I<   sep06 0:00 [kworker/0:0H-events_highpri
1]
root            12  0.0  0.0      0     0 ?        I   sep06 0:00 [kworker/u12:0-ipv6_addrconf
1]
root            13  0.0  0.0      0     0 ?        I<   sep06 0:00 [kworker/R-mm_percpu_wq]
root            14  0.0  0.0      0     0 ?        S    sep06 0:00 [ksoftirqd/0]
root            15  0.0  0.0      0     0 ?        I   sep06 0:00 [rcu_preempt]
root            16  0.0  0.0      0     0 ?        S    sep06 0:00 [rcu_exp_par_gp_kthread_wor
02/0]
root            17  0.0  0.0      0     0 ?        S    sep06 0:00 [rcu_exp_gp_kthread_worker]
root            18  0.0  0.0      0     0 ?        S    sep06 0:00 [migration/0]
root            19  0.0  0.0      0     0 ?        S    sep06 0:00 [idle_inject/0]
root            20  0.0  0.0      0     0 ?        S    sep06 0:00 [cpuhp/0]
root            21  0.0  0.0      0     0 ?        S    sep06 0:00 [cpuhp/1]
```

```
root            8594  0.0  0.0      0     0 ?        I   00:01 0:00 [kworker/u15:1-ext4-rsv-conv
root           12560  0.0  0.0      0     0 ?        I   00:07 0:00 [kworker/0:1-events]
root           12882  0.0  0.0      0     0 ?        I   00:07 0:00 [kworker/u15:0-kvfree_rcu_re
root           13848  0.0  0.0      0     0 ?        I   00:07 0:00 [kworker/2:2-ata_sff]
root           15801  0.0  0.0      0     0 ?        I   00:12 0:00 [kworker/2:0-ata_sff]
root           15892  0.0  0.1 16692 7204 ?        S   00:13 0:00 systemd-userwork: waiting ...
root           15935  0.0  0.0      0     0 ?        I   00:13 0:00 [kworker/1:2]
root           16013  0.0  0.1 16692 7208 ?        S   00:13 0:00 systemd-userwork: waiting ...
root           16083  0.0  0.1 16692 7160 ?        S   00:13 0:00 systemd-userwork: waiting ...
matthew        16413  6.0 11.5 3286444 464468 ?        Sl   00:13 0:09 /usr/lib/firefox-esr/firefox
matthew        16426  0.0  0.0 21012 3340 ?        Sl   00:13 0:00 /usr/lib/firefox-esr/crashsh
matthew        16513  0.0  1.1 240376 46308 ?        Sl   00:13 0:00 /usr/lib/firefox-esr/firefox
pcscd          16535  0.0  0.1 257192 7620 ?        Ssl  00:13 0:00 /usr/sbin/pcscd --foreground
matthew        16546  0.2  3.3 2487952 136404 ?        Sl   00:13 0:00 /usr/lib/firefox-esr/firefox
matthew        16554  0.0  1.4 386776 50004 ?        Sl   00:13 0:00 /usr/lib/firefox-esr/firefox
matthew        16626  0.2  2.9 2479772 117352 ?        Sl   00:13 0:00 /usr/lib/firefox-esr/firefox
matthew        16651  0.4  2.7 2480680 111888 ?        Sl   00:13 0:00 /usr/lib/firefox-esr/firefox
matthew        16710  0.0  1.3 384404 56148 ?        Sl   00:13 0:00 /usr/lib/firefox-esr/firefox
matthew        16723  3.1  5.4 2648020 216748 ?        Sl   00:13 0:04 /usr/lib/firefox-esr/firefox
matthew        16750  1.9  3.9 2527636 159792 ?        Sl   00:13 0:02 /usr/lib/firefox-esr/firefox
matthew        16837  0.1  2.0 2436452 80932 ?        Sl   00:13 0:00 /usr/lib/firefox-esr/firefox
matthew        16905  0.1  2.0 2436452 81184 ?        Sl   00:13 0:00 /usr/lib/firefox-esr/firefox
matthew        16900  0.1  2.0 2436452 81012 ?        Sl   00:13 0:00 /usr/lib/firefox-esr/firefox
root           17305  0.0  0.0      0     0 ?        I   00:14 0:00 [kworker/2:3]
root           17394  0.0  0.0      0     0 ?        I   00:14 0:00 [kworker/0:2-events]
root           17583  0.0  0.0      0     0 ?        I   00:14 0:00 [kworker/u12:2]
root           17844  0.0  0.0      0     0 ?        I   00:15 0:00 [kworker/u15:2-events_unboun
matthew        18338  0.0  0.1 9668 4596 pts/0    R+   00:16 0:00 ps aux
```

```
(matthew@kali-mv)-[~/lab5]
$ ps aux > running_processes.txt
```

```
(matthew@kali-mv)-[~/lab5]
$ ps aux
USER          PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root             1  0.0  0.3 24628 15528 ?        Ss   sep06 0:01 /sbin/init splash
root             2  0.0  0.0      0     0 ?        S    sep06 0:00 [kthreadd]
root             3  0.0  0.0      0     0 ?        S    sep06 0:00 [pool_workqueue_release]
root             4  0.0  0.0      0     0 ?        I<   sep06 0:00 [kworker/R-rcu_gp]
root            10  0.0  0.0      0     0 ?        I<   sep06 0:00 [kworker/R-sync_wq]
root            11  0.0  0.0      0     0 ?        I<   sep06 0:00 [kworker/R-kvfree_rcu_reclai
0]
root             7  0.0  0.0      0     0 ?        I<   sep06 0:00 [kworker/R-slub_flushwq]
root             8  0.0  0.0      0     0 ?        I<   sep06 0:00 [kworker/R-netns]
root            10  0.0  0.0      0     0 ?        I<   sep06 0:00 [kworker/0:0H-events_highpri
1]
root            12  0.0  0.0      0     0 ?        I   sep06 0:00 [kworker/u12:0-ipv6_addrconf
1]
root            13  0.0  0.0      0     0 ?        I<   sep06 0:00 [kworker/R-mm_percpu_wq]
root            14  0.0  0.0      0     0 ?        S    sep06 0:00 [ksoftirqd/0]
root            15  0.0  0.0      0     0 ?        I   sep06 0:00 [rcu_preempt]
root            16  0.0  0.0      0     0 ?        S    sep06 0:00 [rcu_exp_par_gp_kthread_wor
02/0]
root            17  0.0  0.0      0     0 ?        S    sep06 0:00 [rcu_exp_gp_kthread_worker]
root            18  0.0  0.0      0     0 ?        S    sep06 0:00 [migration/0]
root            19  0.0  0.0      0     0 ?        S    sep06 0:00 [idle_inject/0]
root            20  0.0  0.0      0     0 ?        S    sep06 0:00 [cpuhp/0]
root            21  0.0  0.0      0     0 ?        S    sep06 0:00 [cpuhp/1]
root            22  0.0  0.0      0     0 ?        S    sep06 0:00 [idle_inject/1]
root            23  0.0  0.0      0     0 ?        S    sep06 0:00 [migration/1]
root            24  0.0  0.0      0     0 ?        S    sep06 0:00 [ksoftirqd/1]
root            26  0.0  0.0      0     0 ?        I<   sep06 0:00 [kworker/1:0H-events_highpri
1]
root            27  0.0  0.0      0     0 ?        S    sep06 0:00 [cpuhp/2]
root            28  0.0  0.0      0     0 ?        S    sep06 0:00 [idle_inject/2]
root            29  0.0  0.0      0     0 ?        S    sep06 0:00 [migration/2]
root            30  0.0  0.0      0     0 ?        S    sep06 0:00 [ksoftirqd/2]
```

5. Active Network Connection Baseline

Active listening network sockets were collected with ss. The command was also executed with elevated privileges to include available process information, and the resulting socket baseline was saved to active_connections.txt. This provides a reference for identifying unexpected listening ports or services during an alert investigation.

```
(matthew@kali-mv)-[~/lab5]
$ ss -tulpn
Netid  State  Recv-Q  Send-Q  Local Address:Port  Peer Address:Port  Process
tcp    LISTEN  0        4096    127.0.0.1:43151     0.0.0.0:*
```

```
(matthew@kali-mv)-[~/lab5]
$ sudo ss -tulpn
[sudo] contraseña para matthew:
[sudo] State  Recv-Q  Send-Q  Local Address:Port  Peer Address:Port
Process
tcp    LISTEN  0        4096    127.0.0.1:43151     0.0.0.0:*
users:((("containerd",pid=744,fd=12))
```

```
(matthew@kali-mv)-[~/lab5]
$ sudo ss -tulpn > active_connections.txt
```

6. Established TCP Connection Baseline

Current TCP connection state was recorded with ss -tn. Established sessions visible at the time of collection were captured and the output was saved to tcp_connections.txt for later comparison with network activity observed during an investigation.

```
(matthew@kali-mv)-[~/lab5]
$ sudo ss -tn
State  Recv-Q  Send-Q  Local Address:Port  Peer Address:Port
ESTAB  0        0        10.0.2.15:60990     34.107.243.93:443
ESTAB  0        0        10.0.2.15:58778     142.251.152.119:443
```

```
(matthew@kali-mv)-[~/lab5]
$ ss -tn > tcp_connections.txt
```

7. Running Services Baseline

Currently running system services were collected with `systemctl --type=service --state=running`. The active service list was reviewed and saved to `running_services.txt`, providing a known-good reference for detecting unexpected service execution or service-state changes.

```
(matthew@kali-mv)-[~/lab5]
$ systemctl --type=service --state=running
```

UNIT	LOAD	ACTIVE	SUB	DESCRIPTION
accounts-daemon.service	loaded	active	running	Accounts Service
colord.service	loaded	active	running	Manage, Install
containerd.service	loaded	active	running	containerd conta
cron.service	loaded	active	running	Regular backgrou
dbus.service	loaded	active	running	D-Bus System Mes
docker.service	loaded	active	running	Docker Applicati
getty@tty1.service	loaded	active	running	Getty on tty1
haveged.service	loaded	active	running	Entropy Daemon b
lightdm.service	loaded	active	running	Light Display Ma
ModemManager.service	loaded	active	running	Modem Manager
NetworkManager.service	loaded	active	running	Network Manager
pcscd.service	loaded	active	running	PC/SC Smart Card
polkit.service	loaded	active	running	Authorization Ma
rtkit-daemon.service	loaded	active	running	RealtimeKit Sche
systemd-journald.service	loaded	active	running	Journal Service
systemd-logind.service	loaded	active	running	User Login Manag
systemd-udev.service	loaded	active	running	Rule-based Manag
systemd-userdbd.service	loaded	active	running	User Database Ma
udisks2.service	loaded	active	running	Disk Manager
upower.service	loaded	active	running	Daemon for power
user@1000.service	loaded	active	running	User Manager for
virtualbox-guest-utils.service	loaded	active	running	Virtualbox guest

```
Legend: LOAD    → Reflects whether the unit definition was properly load
ACTIVE → The high-level unit activation state, i.e. generalizat
SUB    → The low-level unit activation state, values depend on
```

```
(matthew@kali-mv)-[~/lab5]
$ systemctl --type=service --state=running > running_services.txt
```

8. Enabled Startup Services Baseline

Services configured to start automatically were collected with `systemctl list-unit-files --type=service --state=enabled`. The enabled startup configuration was saved to `startup_services.txt` so that future analysis can identify persistence or configuration changes that were not present in the original baseline.

```
(matthew@kali-mv)-[~/lab5]
$ systemctl list-unit-files --type=service --state=enabled
```

UNIT	FILE	STATE	PRESET
accounts-daemon.service		enabled	enable
console-setup.service		enabled	enable
cron.service		enabled	enable
docker.service		enabled	enable
getty@.service		enabled	enable
haveged.service		enabled	enable
keyboard-setup.service	portA	enabled	enable
lightdm.service		enabled	disabl
ModemManager.service		enabled	enable
networking.service		enabled	enable
NetworkManager-dispatcher.service		enabled	disabl
NetworkManager-wait-online.service		enabled	disabl
NetworkManager.service		enabled	enable
regenerate-ssh-host-keys.service		enabled	enable
smartmontools.service		enabled	enable
systemd-pstore.service		enabled	enable
virtualbox-guest-utils.service		enabled	enable

17 unit files listed.

```
(matthew@kali-mv)-[~/lab5]
$ systemctl list-unit-files --type=service --state=enabled > startup_services.txt
```

9. Consolidated Baseline File

The hostname, current user, running processes, active network connections, running services, and enabled startup services were combined into a single baseline_lab5.txt file. The file creation was verified and its contents were reviewed to confirm that the endpoint state had been preserved in one reference artifact.

```
(matthew@kali-mv)-[~/lab5]
$ {
echo "==== HOSTNAME ===="
hostnamectl

echo "==== CURRENT USER ===="
whoami
id

echo "==== RUNNING PROCESSES ===="
ps aux

echo "==== ACTIVE NETWORK CONNECTIONS ===="
sudo ss -tulnp

echo "==== RUNNING SERVICES ===="
systemctl --type=service --state=running

echo "==== ENABLED STARTUP SERVICES ===="
systemctl list-unit-files --type=service --state=enabled
} > baseline_lab5.txt
```

```
(matthew@kali-mv)-[~/lab5]
$ ls -lh baseline_lab5.txt
-rw-rw-r-- 1 matthew matthew 29K sep  7 00:35 baseline_lab5.txt
```

```
(matthew@kali-mv)-[~/lab5]
$ cat baseline_lab5.txt
==== HOSTNAME ====
Static Hostname: kali-mv
Icon name: computer-vm
Chassis: vm
Machine ID: 7cf819a2bd1546369daa029d730d8270
Boot ID: 29f3efbbe90f4d1f9ef7d3002b042d57
Virtualization: oracle
Operating System: Kali GNU/Linux Rolling
Kernel: Linux 6.18.12-kali-amd64
Architecture: x86_64
Hardware Vendor: innotek GmbH
Hardware Model: VirtualBox
Hardware Version: 1.2
Firmware Version: VirtualBox
Firmware Date: Fri 2006-12-01
Firmware Age: 19y 9month 6d

==== CURRENT USER ====
matthew
uid=1000(matthew) gid=1000(matthew) grupp=1000(matthew),4(adm),20(dialout),24(cdrom),25(floppy),27(sudo),29(audio),30(dip),44(video),46(plugdev),100(users),101(netdev),103(scanner),116(blue
tooth),121(lpadmin),124(wireshark),130(vboxsf),131(kaboxer)

==== RUNNING PROCESSES ====
USER      PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root         1  0.1  0.3 24628 15528 ?        Ss   Sep06   0:03 /sbin/init splash
root         2  0.0  0.0      0     0 ?        S    Sep06   0:00 [kthreadd]
root         3  0.0  0.0      0     0 ?        S    Sep06   0:00 [pool_workqueue_release]
root         4  0.0  0.0      0     0 ?        Ic   Sep06   0:00 [kworker/R-rcu_gp]
```

```
matthew@kali-mv:~/lab5
Session Acciones Editar Vista Ayuda

-esr -contentproc -isForBrowser -prefsHandle 0:40357 -prefMapHandle 1:272496 -jsInitHandle 2:24
2716 -parentBuildID 20268317093326 -sandboxReporter 3 -chrootClient 4 -ipcHandle 5 -initialChan
nelID {f0a2f82-9806-4256-aa8c-5194897abc12} -parentPid 10413 -crashReporter 6 -crashHelper 7 -
greonmi /usr/lib/firefox-esr/omni.ja -appomni /usr/lib/firefox-esr/browser/omni.ja -appDir /usr
/lib/firefox-esr/browser 12 tab
root      17583  0.0  0.0      0     0 ?        I    00:14   0:00 [kworker/u12:2]
root      17844  0.0  0.0      0     0 ?        I    00:15   0:00 [kworker/u12:2-flush-8:0]
root      23218  0.0  0.0      0     0 ?        I    00:25   0:00 [kworker/0:1-events]
root      25269  0.0  0.0      0     0 ?        I    00:29   0:00 [kworker/2:2-ata_sff]
root      27425  0.0  0.1 16552 6776 ?        S    00:33   0:00 systemd-userwork: waiting...
root      27592  0.0  0.1 16552 7184 ?        S    00:33   0:00 systemd-userwork: waiting...
root      27658  0.0  0.1 16552 6780 ?        S    00:34   0:00 systemd-userwork: waiting...
root      27874  0.0  0.0      0     0 ?        I    00:34   0:00 [kworker/2:0-ata_sff]
root      28277  0.0  0.0      0     0 ?        I    00:35   0:00 [kworker/0:2]
root      28538 33.3  0.1 17300 8000 ?        Ss   00:35   0:00 /usr/lib/systemd/systemd-hos
tname
matthew  28543  0.0  0.1 9668 4512 pts/0    R+   00:35   0:00 ps aux

==== ACTIVE NETWORK CONNECTIONS ====
Netid State Recv-Q Send-Q Local Address:Port Peer Address:PortProcess
tcp LISTEN 0 4096 127.0.0.1:43151 0.0.0.0:* users:((("containerd",pid=744,f
d=12))

==== RUNNING SERVICES ====
UNIT                                LOAD    ACTIVE SUB    DESCRIPTION
accounts-daemon.service             loaded active running Accounts Service
colord.service                       loaded active running Manage, Install and Generate Color Profi
les
containerd.service                  loaded active running containerd container runtime
cron.service                         loaded active running Regular background program processing da
```

10. Baseline Artifact Summary

The collected files provide a repeatable system-state reference that can be compared against evidence gathered during future alert investigation. A difference in processes, sockets, TCP sessions, running services, or startup configuration can be reviewed against this initial baseline to determine whether the endpoint state has changed.

Baseline Item	Command	Saved Output
Hostname	hostnamectl	hostname.txt
Current user	whoami / id	current_user.txt
Running processes	ps aux	running_processes.txt
Active network connections	sudo ss -tulpn	active_connections.txt
Established TCP sessions	ss -tn	tcp_connections.txt
Running services	systemctl --type=service --state=running	running_services.txt
Startup services	systemctl list-unit-files --type=service --state=enabled	startup_services.txt
Consolidated baseline	Combined command output	baseline_lab5.txt

11. Conclusions

The endpoint baseline activity was successfully completed on the Kali Linux laboratory system. Host identity, current user information, running processes, active network connections, established TCP sessions, running services, and enabled startup services were collected and saved for later comparison. A consolidated baseline file was also created, providing a practical reference that can support alert investigation by highlighting changes in endpoint processes, network activity, and service configuration.